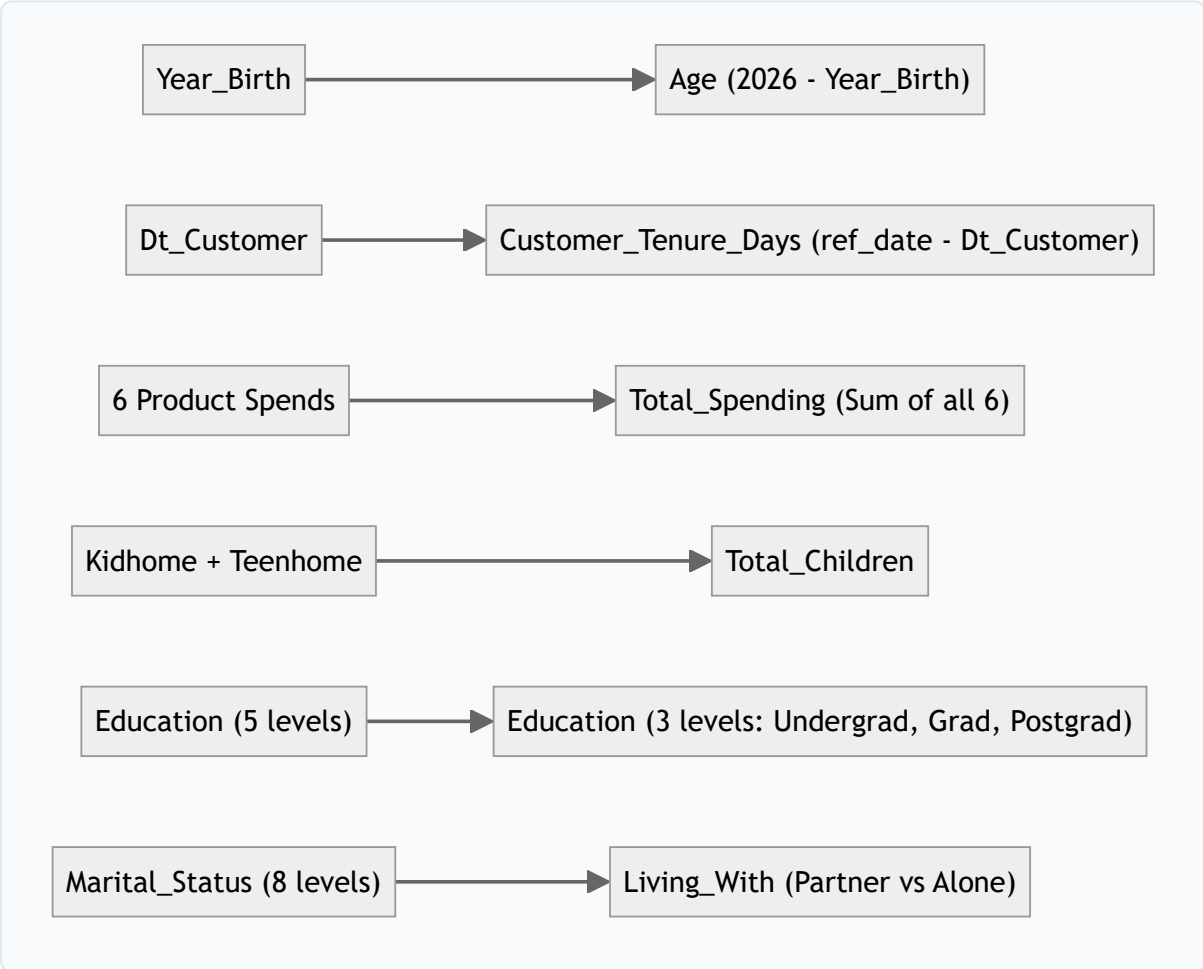
Submodule 33.03: Feature Engineering (Domain-Specific Transformations)

Topic Overview & Theoretical Motivation

Feature engineering is the single most decisive step in unsupervised machine learning. Clustering algorithms cannot autonomously synthesize higher-order commercial concepts; they merely calculate distances between raw vector coordinates.

In this module, raw attributes are converted into high-signal behavioral and demographic variables.



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1. Feature Synthesis Derivations

A. Age Synthesis

Raw birth years (e.g., 1957, 1984) distort clustering geometry. Age provides direct commercial interpretability:

$$\text{Age}_i = 2026 - \text{Year_Birth}_i$$

B. Customer Loyalty Tenure (`Customer_Tenure_Days`)

Clustering distance formulas cannot process date strings (`"2012-09-04"`).

- **Critical Engineering Practice**: Rather than using `datetime.now()` (which shifts dynamically and destroys pipeline reproducibility), we anchor against the maximum joining date in the dataset:

$$\text{Reference Date} = \max(\text{Dt_Customer})$$

$$\text{Customer_Tenure_Days}_i = (\text{Reference Date} - \text{Dt_Customer}_i)$$

C. Total Customer Spending (`Total_Spending`)

Our objective is **Customer Segmentation**, not Product Basket Analysis. We sum all six individual product expenditure categories:

$$\text{Total_Spending} = \text{MntWines} + \text{MntFruits} + \text{MntMeatProducts} + \text{MntFishProc}$$

D. Household Dependency (`Total_Children`)

Consolidating young children and teenagers into a single household dependency count:

$$\text{Total_Children} = \text{Kidhome} + \text{Teenhome}$$

E. Education Level Consolidation

The raw `Education` attribute contains 5 granular categories. We consolidate them into 3 ordinal tiers:

- `Basic`, `2n Cycle` → `Undergraduate` (pursuing or basic education)
- `Graduation` → `Graduate` (bachelor's completed)
- `Master`, `PhD` → `Postgraduate` (advanced post-grad degrees)

F. Living Status Consolidation (`Living_With`)

The raw `Marital_Status` column contains 8 categories (`Married`, `Together`, `Single`, `Divorced`, `Widow`, `Alone`, `Absurd`, `YOLO`). For retail basket dynamics, the key factor is whether the customer shares household expenses with a partner or lives alone:

- `Married`, `Together` → `Partner`
- `Single`, `Divorced`, `Widow`, `Alone`, `Absurd`, `YOLO` → `Alone`

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2. Python Implementation

Reference: [smartcart.ipynb] (Cells 09 - 15).

```
# 1. Age
df["Age"] = 2026 - df["Year_Birth"]

# 2. Customer Joining Date & Tenure
df["Dt_Customer"] = pd.to_datetime(df["Dt_Customer"], dayfirst=True)
reference_date = df["Dt_Customer"].max()
df["Customer_Tenure_Days"] = (reference_date - df["Dt_Customer"]).dt.days

# 3. Total Spending
df["Total_Spending"] = (
    df["MntWines"] + df["MntFruits"] + df["MntMeatProducts"] +
    df["MntFishProducts"] + df["MntSweetProducts"] + df["MntGoldProds"]
)

# 4. Total Children
df["Total_Children"] = df["Kidhome"] + df["Teenhome"]

# 5. Education Mapping
df["Education"] = df["Education"].replace({
    "Basic": "Undergraduate", "2n Cycle": "Undergraduate",
    "Graduation": "Graduate",
    "Master": "Postgraduate", "PhD": "Postgraduate"
})

# 6. Marital Status Mapping -> Living_With
df["Living_With"] = df["Marital_Status"].replace({
    "Married": "Partner", "Together": "Partner",
    "Single": "Alone", "Divorced": "Alone",
```

```
    "Widow": "Alone", "Absurd": "Alone", "YOLO": "Alone"
})
```